

# Solar Feasibility Proposal

6.0 kWp · 15 panels · 9502 kWh per year

Prepared by solarVis AI · 29 July 2026 · Reference VtkG70YaclLw

## Headline results

SYSTEM SIZE

**6.0 kWp**

15 of 15 panels requested

ANNUAL PRODUCTION

**9,502 kWh**

68.9% of your usage

ANNUAL SAVING

**€2,376**

at €0.25/kWh

PAYBACK

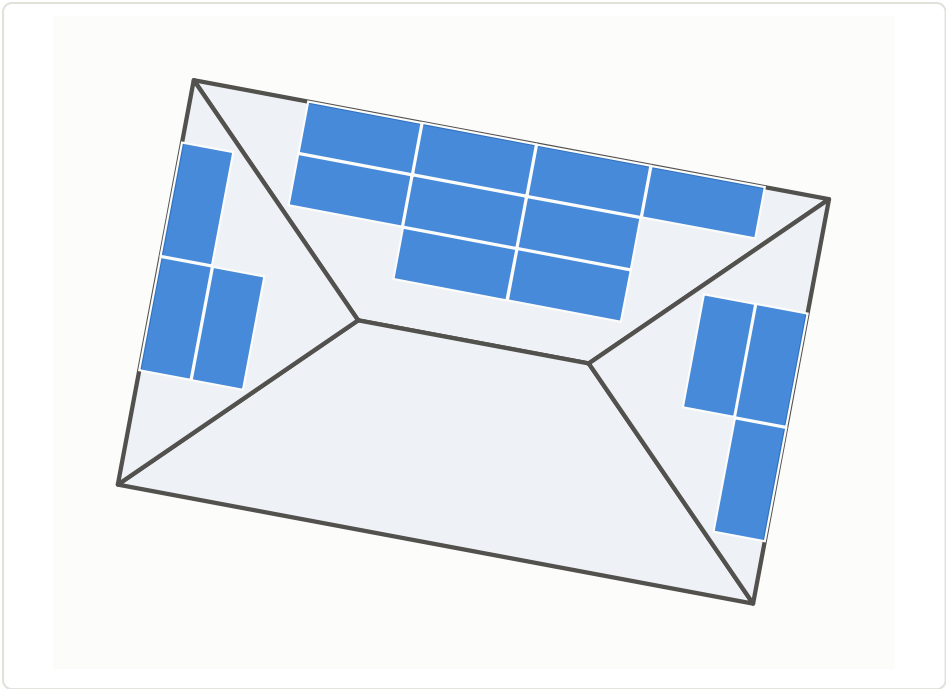
**3.7 yr**

20-year net €38,721

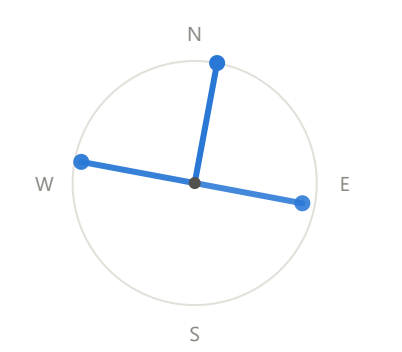
## Project inputs

|                        |                                       |
|------------------------|---------------------------------------|
| Location as entered    | -34.04658242871865, 18.46491476666948 |
| Location analysed      | -34.046582, 18.464915                 |
| Monthly consumption    | 1,150 kWh                             |
| Annual consumption     | 13,800 kWh                            |
| Electricity unit price | €0.25 / kWh                           |
| Requested system size  | 6.0 kWp (15 panels)                   |
| Feasible system size   | 6.0 kWp (15 panels)                   |

Roof reconstruction



Orientation



Ray length and weight show relative yield. This site is in the southern hemisphere, so north-facing roof faces produce most.

Plan view of the reconstructed roof and panel layout, drawn to scale from the measured geometry.

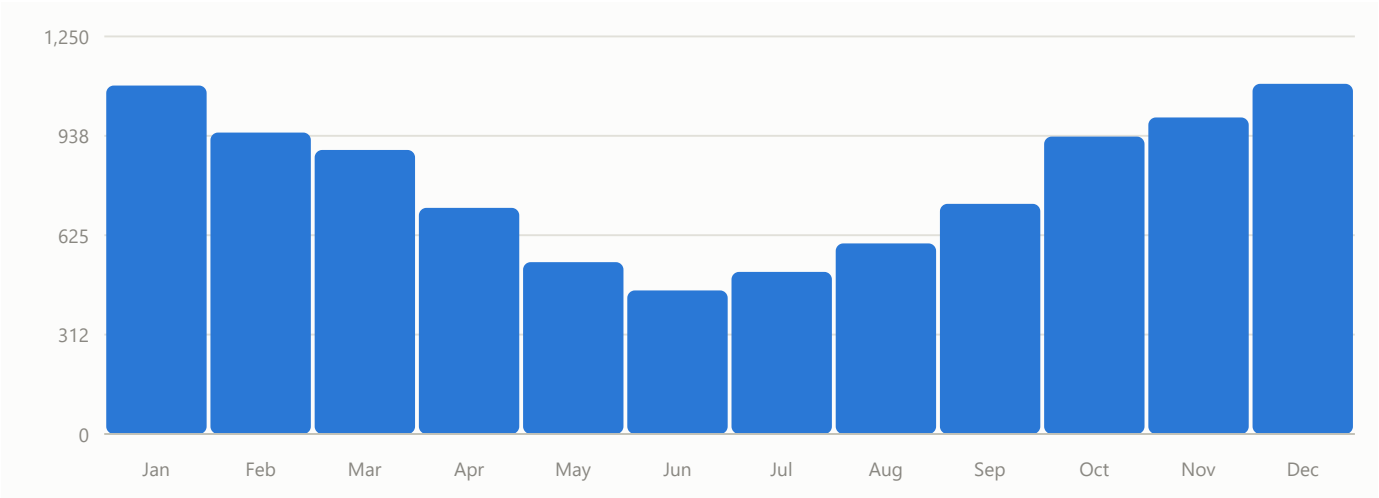
| Facet       | Azimuth | Pitch | Plan area            | Sloped area          | Panels |
|-------------|---------|-------|----------------------|----------------------|--------|
| North facet | 10.6°   | 25°   | 27.30 m <sup>2</sup> | 30.12 m <sup>2</sup> | 9      |
| South facet | 190.6°  | 25°   | 27.30 m <sup>2</sup> | 30.12 m <sup>2</sup> | 0      |
| West facet  | 280.6°  | 25°   | 12.76 m <sup>2</sup> | 14.08 m <sup>2</sup> | 3      |
| East facet  | 100.6°  | 25°   | 12.76 m <sup>2</sup> | 14.07 m <sup>2</sup> | 3      |
| Total       |         |       | 80.12 m <sup>2</sup> | 88.40 m <sup>2</sup> | 15     |

Edge measurements

| Edge    | Type  | Plan length | True length |
|---------|-------|-------------|-------------|
| eave_0  | eave  | 11.216 m    | 11.216 m    |
| eave_1  | eave  | 7.143 m     | 7.143 m     |
| eave_2  | eave  | 11.216 m    | 11.216 m    |
| eave_3  | eave  | 7.143 m     | 7.143 m     |
| hip_0   | hip   | 5.051 m     | 5.319 m     |
| hip_1   | hip   | 5.051 m     | 5.318 m     |
| hip_2   | hip   | 5.051 m     | 5.318 m     |
| hip_3   | hip   | 5.051 m     | 5.318 m     |
| ridge_0 | ridge | 4.073 m     | 4.073 m     |

Lengths are measured on the source satellite raster at 0.061850 m per pixel, derived from the verified Web Mercator zoom and scale. Hip lengths use true 3-D endpoint geometry; they are deliberately not the plan length divided by cos(pitch), which applies only to an edge running straight up the slope.

Energy production



Modelled monthly production across all occupied roof facets.

| Facet       | Panels | Capacity | PVGIS aspect | Specific yield | Annual production |
|-------------|--------|----------|--------------|----------------|-------------------|
| North facet | 9      | 3.6 kWp  | -169.4°      | 1,679 kWh/kWp  | 6,043 kWh         |
| West facet  | 3      | 1.2 kWp  | 100.6°       | 1,515 kWh/kWp  | 1,818 kWh         |
| East facet  | 3      | 1.2 kWp  | -79.4°       | 1,367 kWh/kWp  | 1,641 kWh         |
| Total       | 15     | 6.0 kWp  |              |                | 9,502 kWh         |

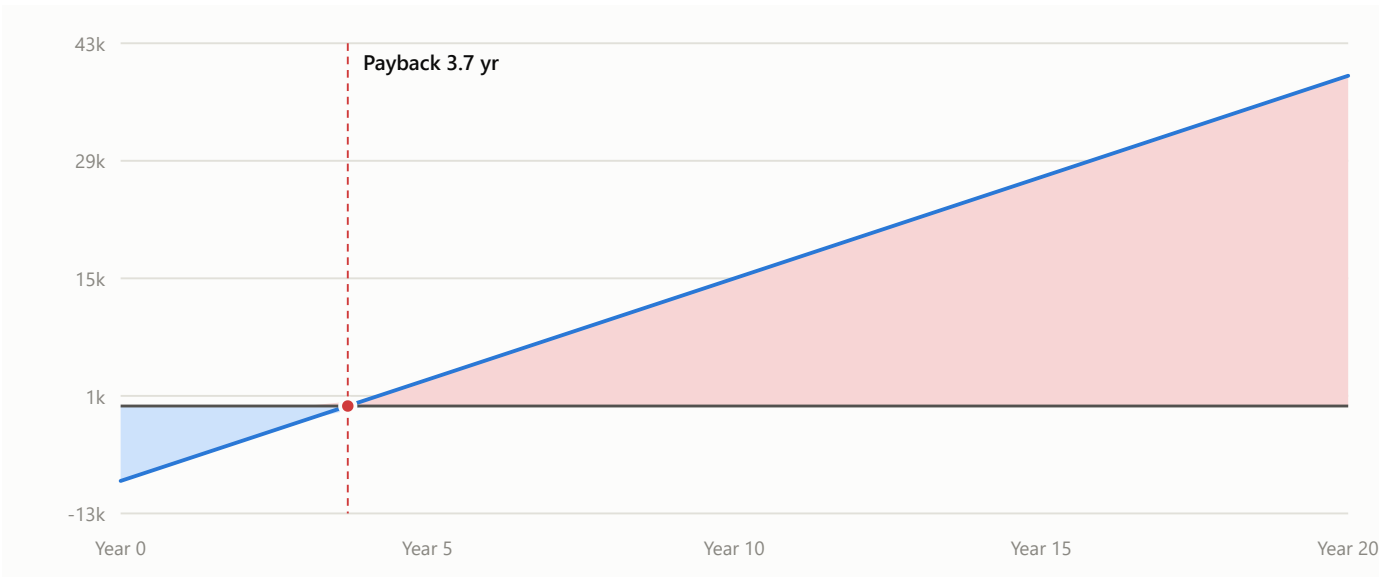
Source: PVGIS 5.3 (PVGIS-SARAH3), European Commission Joint Research Centre. One normalised 1 kWp request per roof facet, using that facet's own pitch and azimuth; production is that yield times the capacity actually placed. Data mode: **live**. Retrieved from [https://re.jrc.ec.europa.eu/api/v5\\_3/PVcalc](https://re.jrc.ec.europa.eu/api/v5_3/PVcalc) at 2026-07-29T19:25:26.145138+00:00.

Financial analysis

|                                        |                     |
|----------------------------------------|---------------------|
| Annual production                      | 9,502 kWh           |
| Annual consumption                     | 13,800 kWh          |
| Energy covered (capped at consumption) | 9,502 kWh           |
| Consumption coverage                   | 68.9%               |
| Annual saving                          | €2375.55            |
| Capital cost as quoted                 | \$10000.00 USD      |
| USD/EUR reference rate applied         | 0.87897             |
| Rate date                              | 2026-07-24          |
| Rate provider                          | ECB via Frankfurter |
| Capital cost used in this analysis     | €8789.70 EUR        |
| Simple payback                         | 3.70 years          |
| Net benefit over 20 years              | €38721.30           |

Reference rate source: **development fixture rate — not live market data**. The capital cost is quoted in US dollars while savings accrue in euro, so the two are never compared directly. This rate was fixed when the proposal was created and is stored with it: reopening this proposal later will not recalculate these figures at a newer rate.

Twenty-year cash flow



Cumulative position. Below the line the system is still repaying its capital cost; above it, it is returning value.

| Year | Saving   | Cumulative | Year | Saving   | Cumulative |
|------|----------|------------|------|----------|------------|
| 0    | €0.00    | €-8789.70  | 11   | €2375.55 | €17341.35  |
| 1    | €2375.55 | €-6414.15  | 12   | €2375.55 | €19716.90  |
| 2    | €2375.55 | €-4038.60  | 13   | €2375.55 | €22092.45  |
| 3    | €2375.55 | €-1663.05  | 14   | €2375.55 | €24468.00  |
| 4    | €2375.55 | €712.50    | 15   | €2375.55 | €26843.55  |
| 5    | €2375.55 | €3088.05   | 16   | €2375.55 | €29219.10  |
| 6    | €2375.55 | €5463.60   | 17   | €2375.55 | €31594.65  |
| 7    | €2375.55 | €7839.15   | 18   | €2375.55 | €33970.20  |
| 8    | €2375.55 | €10214.70  | 19   | €2375.55 | €36345.75  |
| 9    | €2375.55 | €12590.25  | 20   | €2375.55 | €38721.30  |
| 10   | €2375.55 | €14965.80  |      |          |            |

Assumptions and limitations

- Fixed case property; geocoding is out of scope, so any entered location resolves to it.
  - The brief prints the latitude without a minus sign, which falls in open sea. The southern-hemisphere reading is used; both values are on record.
  - Uniform roof pitch of 25° across all four facets.
  - Panels are 1 m x 2 m, 400 Wp, flush-mounted in the plane of the roof.
  - PVGIS system loss 14%, crystSi, building-mounted.
  - Electricity price held flat at €0.25/kWh.
  - Savings are capped at consumption: no export compensation is assumed.
- No module degradation, inflation, maintenance cost, financing cost or tax effect.
  - No shading analysis and no detection of chimneys, vents or other obstructions.
  - Roof-edge setback 0 m; inter-panel gap 0.02 m.
  - Imagery date is unknown and may predate recent changes to the property.
  - **The USD/EUR rate in this proposal is development fixture data, not a live market rate.**
  - **Satellite imagery is a development fixture rendered on the verified map grid, not a live Google Static Maps request.**

This is a feasibility estimate produced from satellite imagery and modelled irradiation data, not a site survey or a binding quotation. Production and financial outcomes will vary with weather, shading, equipment, tariff changes and installation detail. A physical survey should precede any purchase.